

Android Application Development

Important Exam Questions — 20 Must-Know Questions

Covers all 5 units · Suitable for internal tests, mid-semester, and end-semester examinations

Unit 1 — Introduction to Android

- Q1.** Explain the concept of a Mobile Operating System. Compare Android and iOS in terms of architecture, openness, security, app distribution, and customization. Which platform dominates globally and why?
- Q2.** What are the prominent features of Android as a platform? Trace the evolution of Android versions, highlighting major feature additions at each significant release (mention at least five versions).
- Q3.** Describe the Android Studio IDE in detail. What are the steps to create a new Android project, configure an AVD (Android Virtual Device), and run the app on an emulator? How does running on a physical device differ?

Unit 2 — Android Architecture & Components

- Q4.** Draw and explain the complete Android Architecture with all five layers — Linux Kernel, Libraries, Android Runtime, Application Framework, and Applications. What specific functions does each layer perform?
- Q5.** What are the four Application Components of Android? Explain the complete lifecycle of an Activity with all callback methods, and contrast it with the lifecycle of a Service (started vs. bound service).
- Q6.** What is a Broadcast Receiver in Android? How does it listen for system-wide broadcasts? Give two examples of system broadcasts. How does a Content Provider differ, and what role does it play in data sharing between apps?
- Q7.** What is the AndroidManifest.xml file? List and explain at least six things that must be declared in it. What is the significance of permissions in the Manifest, and what happens if an Activity is not registered there?

Unit 3 — UI Design, Widgets & Event Handling

- Q8.** Explain all four layout types in Android: `LinearLayout`, `RelativeLayout`, `ConstraintLayout`, and `GridLayout`. Compare their XML attributes and explain which layout is best suited for complex, responsive UI design and why.
- Q9.** Describe the following UI widgets and their important XML attributes: `TextView`, `EditText`, `Button`, `ImageView`, and `ScrollView`. How do you reference them in Java/Kotlin using `findViewById()`?
- Q10.** What is `RecyclerView` and how does it improve upon `ListView`? Explain the roles of `RecyclerView.Adapter`, `ViewHolder` pattern, and `LayoutManager`. Why is `RecyclerView` more memory-efficient for large data sets?
- Q11.** Explain Event-driven Programming in Android. Describe the four event handling methods — `onClick()`, `onKey()`, `onTouch()`, and `onFocusChange()` — and explain the concept of `onItemSelectedListener` for `ListView` item selection.
- Q12.** What is the difference between Explicit and Implicit Intents? Explain how to pass data between two Activities using `Intent`. What is an Intent Filter, and how does Android resolve which Activity handles an Implicit Intent?
- Q13.** Explain Toasts and AlertsDialogs in Android with their creation steps. Describe the three menu types — Options Menu, Context Menu, and Popup Menu — including how each is inflated and handled.

Unit 4 — Data Storage

- Q14.** What are Shared Preferences? Explain how to write and read data using `SharedPreferences` with `getSharedPreferences()` and `SharedPreferences.Editor`. What are the limitations of Shared Preferences?
- Q15.** Explain SQLite Database in Android. What is `SQLiteOpenHelper` and what methods does it provide? Describe all four CRUD operations — Insert, Read (query), Update, and Delete — with the methods and parameters involved.
- Q16.** What is Firebase Realtime Database? How do you integrate Firebase into an Android Studio project? Explain the steps to Insert data into Firebase, Retrieve data using a `ValueEventListener`, and display it in the UI.

Unit 5 — Kotlin, Cross-Platform, APIs & Publishing

- Q17.** What is Kotlin and why is it now the preferred language for Android development? List at least five features of Kotlin (e.g., null safety, extension functions, data classes) and contrast them with equivalent Java concepts.
- Q18.** Compare Flutter and React Native as cross-platform mobile development frameworks. Discuss their language, performance, UI rendering approach, community support, and suitability for enterprise applications.
- Q19.** What is an API? Explain the process of consuming a REST API in Android — from making an HTTP request to parsing a JSON response. What libraries are commonly used for API calls and JSON parsing in Android?
- Q20.** What are the steps involved in publishing an Android application on the Google Play Store? Explain the significance of: signing the APK/AAB, setting up a Play Console account, configuring the app listing, and the review process.
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— End of Exam Question Bank —